higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

INSTRUMENT TECHNOLOGY
NQF LEVEL 2

(6060452)

18 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 5 pages.

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TIME: 3 HOURS
MARKS: 150

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Use only BLUE or BLACK ink.
6. Write neatly and legibly.

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QUESTION 1

1.1 During process operations, there are many factors that need to be monitored and controlled to ensure that the process runs smoothly and that the end product conforms to specifications.

1.1.1 Based on the statement above, what does process instrumentation refer to? (3)

1.1.2 List FOUR variables that can be monitored in a process plant. (4)

1.1.3 Name the system that is used to make sure that the process plant operates optimally. (1)

1.2 Complete the table below:
(List only one.)

GROUP	HAZARDOUS MATERIAL IN SURROUNDING ATMOSPHERE
Group A	1.2.1
Group B	1.2.2
Group C	1.2.3
Group G	1.2.4

(4)

1.3 Explain the general functions of an instrument mechanician. (3)

[15]

QUESTION 2

2.1 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (2.1.1-2.1.5) in the ANSWER BOOK.

2.1.1 When a car door is open, a light usually switches on in the interior. Switches in a car that determine whether the car doors are open or closed are examples of limit switches.

2.1.2 A positioner is a device that controls air pressure.

2.1.3 Not all cushion-loaded actuators work in the same way or have the same components to create a cushion pressure.

2.1.4 Stiction is a condition whereby a component has difficulty getting into motion.

2.1.5 Viscosity is the state of being thick, sticky and semi-fluid in consistency.

(5 × 1) (5)

2.2 Explain the working principle of balance actuators. (3)

2.3 General maintenance requirements:

2.3.1 List THREE types of maintenance and explain when it is necessary. (6)

2.3.2 Give FOUR general maintenance requirements of control valves. (4)

2.3.3 Name and explain each of the FOUR specific steps that must be followed to carry out maintenance on control valves. (12)

2.4 Name the THREE main types of control valve actuator problems that commonly occur in process systems. (3)

2.5 What are the FIVE questions an instrument mechanician should ask when inspecting valve systems? (5)
[38]

QUESTION 3

3.1 Manometers

3.1.1 List THREE applications of manometers. (3)

3.1.2 When are inclined manometers useful and why? (3)

3.1.3 Manometers are useful pressure measurement instruments, but they have limitations.

List FOUR of their main limitations. (8)

3.1.4 It is sometimes necessary to clean the manometers that are used.

Explain the guidelines that must be followed when cleaning manometers. (5)

3.2 Make a neat labeled sketch of a C-type Bourdon tube. (8)

3.3 Explain Charles's law. (3)
[30]

QUESTION 4

4.1 The principle of temperature measurements.

4.1.1 Explain the theory of resistance. (4)

4.1.2 Define thermometry. (2)

4.2 Thermocouples

4.2.1 Make a drawing to show the components of a thermocouple. (5)

4.2.2 What are the general steps that must be followed when installing thermocouples? (5)

4.2.3 Explain how you will go about performing the following tasks on a thermocouple: zero-checking, calibrating and verifying. (6)

4.2.4 How do you commission thermocouples? (3)

4.3 What are the critical factors you must consider when using an RTD calibrator to calibrate an RTD? (5)

[30]

QUESTION 5

5.1 Calculate the amount of energy that the fluid possesses when fluid is flowing from a 5 m high ledge into a container at a speed of 5 m/s, at 150 Pa pressure. (6)

5.2 Name FIVE causes for a flowmeter to have inaccurate control. (5)

5.3 Make drawings to show the working of the following instruments:

5.3.1 rotameter

5.3.2 v-cone

5.3.3 Dall-tube

5.3.4 Orifice plate

5.3.5 Segmental wedge

(5 × 5) (25)

5.4 Complete the following sentence by filling the missing word. Write only the word next to the question number (5.4) in the ANSWER BOOK.

… is when the flow of fluid at any point has a velocity that changes all the time so that the flow is not steady and even. (1)

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TOTAL: 150

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